

Section 14.4: Tangent Planes and Linearization

Partial Derivatives

MTH 310
Calculus III

Your Turn: Derivatives Warm-Up

We'll need the partial derivatives of the **hemi-ellipsoid**

$$f(x, y) = \sqrt{25 - 4x^2 - y^2} = (25 - 4x^2 - y^2)^{1/2}$$

Compute the partial derivatives f_x and f_y , then evaluate each at $(2, 1)$.

Hint: Rewrite as a $\frac{1}{2}$ -power so you can use the **chain rule**.

Finding f_x : Hold y constant, chain rule on the $\frac{1}{2}$ -power:

$$f_x = \frac{1}{2} (25 - 4x^2 - y^2)^{-1/2} \cdot (-8x) = \frac{-4x}{\sqrt{25 - 4x^2 - y^2}}$$

Finding f_y : Hold x constant, same chain rule pattern:

$$f_y = \frac{1}{2} (25 - 4x^2 - y^2)^{-1/2} \cdot (-2y) = \frac{-y}{\sqrt{25 - 4x^2 - y^2}}$$

Evaluate at $(2, 1)$: The denominator is $\sqrt{25 - 16 - 1} = \sqrt{8} = 2\sqrt{2}$

$$f_x(2, 1) = \frac{-8}{2\sqrt{2}} = -2\sqrt{2} \approx -2.83 \quad f_y(2, 1) = \frac{-1}{2\sqrt{2}} = -\frac{\sqrt{2}}{4} \approx -0.35$$

The chain rule is the key skill here — every square root, exponential, or composition in f will need it for partial derivatives.

From Tangent Lines to Tangent Planes

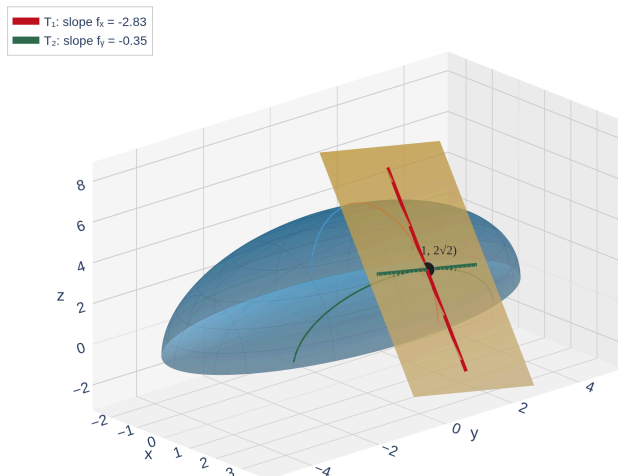
Recall from Calc I: If $f(x)$ is differentiable, then when you zoom in near a point the graph looks like a straight line — the **tangent line**.

The tangent line at x_0 is the best **linear approximation**:

$$y - y_0 = f'(x_0)(x - x_0)$$

Now in Calc III: For a function of two variables $z = f(x, y)$, zooming in near a point makes the surface look like a **flat plane** — the **tangent plane**.

The tangent plane is the best linear approximation to the surface near that point.



[Interactive version](#)

Calc I: zoom in → tangent **line**. Calc III: zoom in → tangent **plane**.

The Tangent Plane Equation

In Calc I we needed one slope $f'(x_0)$. Now we have **two** independent directions to tilt — one for x and one for y — so we need **both partial derivatives**.

Equation of the Tangent Plane

If f is differentiable at (x_0, y_0) , the tangent plane to $z = f(x, y)$ at the point (x_0, y_0, z_0) is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

where $z_0 = f(x_0, y_0)$.

	Calc I (Tangent Line)	Calc III (Tangent Plane)
Function	$y = f(x)$	$z = f(x, y)$
Slopes needed	$f'(x_0)$	$f_x(x_0, y_0)$ and $f_y(x_0, y_0)$
Equation	$y - y_0 = f'(x_0)(x - x_0)$	$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

Same structure as the tangent line equation, but with **two slope terms** instead of one.

Common mistake: Students write the tangent plane formula but forget to **evaluate everything at the point**. The function value $f(x_0, y_0)$ and the partial derivatives $f_x(x_0, y_0)$, $f_y(x_0, y_0)$ must all be computed as **numbers**. Your final answer should be the equation of a line (Calc I) or a plane (Calc III) — no derivatives or function values left!

When Does the Tangent Plane Exist?

Warning: For the tangent plane to exist, we need the function to be **differentiable**. In Calc I, if $f'(x_0)$ exists, then f is differentiable at x_0 . For two variables, it is **not enough** for the partial derivatives to merely exist at the point.

Theorem: Differentiability (Sufficient Condition)

If the partial derivatives f_x and f_y are **continuous** on an open region containing (x_0, y_0) , then f is differentiable at (x_0, y_0) .

In other words: continuous partials $\Rightarrow$ differentiable $\Rightarrow$ tangent plane exists.

Why is "partials exist" not enough?

A function can have partial derivatives at a point yet still have a sharp crease or corner there. Continuous partials **near** the point guarantee the surface is truly smooth.

Rule of thumb: If f_x and f_y are built from standard functions (polynomials, trig, exponentials, etc.) and the point is in the domain, you're safe.

Example 1: Does the Tangent Plane Exist?

Example 1

Does the hemi-ellipsoid $f(x, y) = \sqrt{25 - 4x^2 - y^2}$ have a tangent plane at $(2, 1)$? Why?

From the warm-up, the partial derivatives are:

$$f_x(x, y) = \frac{-4x}{\sqrt{25 - 4x^2 - y^2}}, \quad f_y(x, y) = \frac{-y}{\sqrt{25 - 4x^2 - y^2}}$$

These are built from standard functions (quotients of polynomials and square roots) and are **continuous** wherever the denominator $\sqrt{25 - 4x^2 - y^2} \neq 0$ — that is, everywhere in the interior of the domain.

At $(2, 1)$: $\sqrt{25 - 16 - 1} = 2\sqrt{2} \neq 0$, so $(2, 1)$ is an interior point.

Yes — f_x and f_y are continuous near $(2, 1)$, so f is differentiable there and the tangent plane exists.

Example 2: Tangent Plane to a Hemi-Ellipsoid

Example 2

Find the equation of the tangent plane to the surface

$$z = f(x, y) = \sqrt{25 - 4x^2 - y^2}$$

at the point $(2, 1)$.

Find $f(2, 1)$, $f_x(2, 1)$, $f_y(2, 1)$, and use: $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

Step 1 — Gather the ingredients: From the warm-up,

$$z_0 = f(2, 1) = 2\sqrt{2}, \quad f_x(2, 1) = -2\sqrt{2}, \quad f_y(2, 1) = -\frac{\sqrt{2}}{4}$$

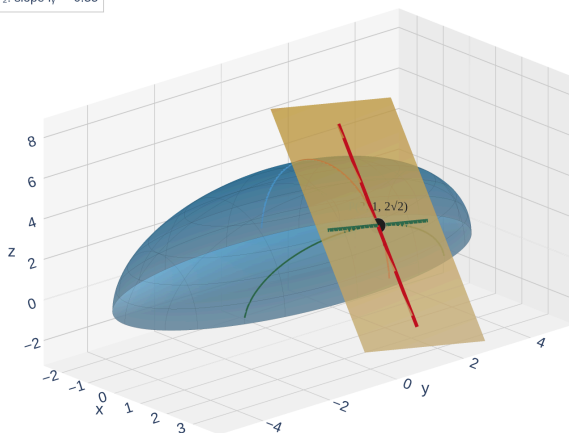
Step 2 — Write the tangent plane formula:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Step 3 — Substitute:

$$z - 2\sqrt{2} = -2\sqrt{2}(x - 2) - \frac{\sqrt{2}}{4}(y - 1)$$

T₁: slope f_x = -2.83
T₂: slope f_y = -0.35



[Interactive version](#)

The tangent plane tilts steeply in the x -direction (slope $-2\sqrt{2} \approx -2.83$) but gently in y (slope $-\sqrt{2}/4 \approx -0.35$).

Your Turn: Find a Tangent Plane

Example 3

Find the equation of the tangent plane to

$$z = x^2 + y^2$$

at the point $(1, 2, 5)$.

Compute f_x and f_y , evaluate at $(1, 2)$, and plug into the formula.

Gather the ingredients:

$$f(1, 2) = 5, \quad f_x = 2x \Rightarrow f_x(1, 2) = 2, \quad f_y = 2y \Rightarrow f_y(1, 2) = 4$$

Write the formula:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Substitute:

$$z - 5 = 2(x - 1) + 4(y - 2)$$

Simplify:

$$z - 5 = 2x - 2 + 4y - 8 \Rightarrow z = 2x + 4y - 5$$

Quick check: Does $(1, 2, 5)$ satisfy this? $2(1) + 4(2) - 5 = 5 \checkmark$

Linear Approximation

Once we have the tangent plane, we can use it to **estimate** the value of f at nearby points — exactly as we used the tangent line in Calc I.

Linearization

The **linearization** (or **linear approximation**) of f at (a, b) is:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

For (x, y) near (a, b) : $f(x, y) \approx L(x, y)$.

Notice: $L(x, y)$ is just the tangent plane equation solved for z . The tangent plane **is** the linear approximation.

The tangent plane gives us a simple formula to **approximate** a complicated function near a known point.

Example 4: Approximate $f(2.05, 0.92)$

Example 4

Using the tangent plane from Example 2, approximate the value of

$$f(2.05, 0.92) = \sqrt{25 - 4(2.05)^2 - (0.92)^2}$$

Use $L(x, y) = f(2, 1) + f_x(2, 1)(x - 2) + f_y(2, 1)(y - 1)$ with $x = 2.05$, $y = 0.92$.

Write the linearization formula:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

From Example 2: $f(2, 1) = 2\sqrt{2}$, $f_x(2, 1) = -2\sqrt{2}$, $f_y(2, 1) = -\frac{\sqrt{2}}{4}$

Substitute:

$$L(2.05, 0.92) = 2\sqrt{2} + (-2\sqrt{2})(2.05 - 2) + \left(-\frac{\sqrt{2}}{4}\right)(0.92 - 1)$$

Simplify:

$$\begin{aligned} &= 2\sqrt{2} + (-2\sqrt{2})(0.05) + \left(-\frac{\sqrt{2}}{4}\right)(-0.08) \\ &= 2\sqrt{2} - 0.1\sqrt{2} + 0.02\sqrt{2} = 1.92\sqrt{2} \approx 2.7153 \end{aligned}$$

Calculator check: $f(2.05, 0.92) = \sqrt{25 - 16.81 - 0.8464} \approx 2.7109$

Error ≈ 0.004 — very close!

For small changes in x and y , the tangent plane gives an excellent estimate without needing a calculator.

Differentials

There is an equivalent way to think about linear approximation using **differentials** — the same idea from Calc I, extended to two variables.

Total Differential

If $z = f(x, y)$ and f is differentiable, the **total differential** is:

$$dz = f_x(x, y) dx + f_y(x, y) dy$$

where $dx = \Delta x$ and $dy = \Delta y$ are the changes in x and y .

Connection to linear approximation:

- dz is the change predicted by the tangent plane (the approximation)

- $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$ is the actual change
- For small dx and dy : $\Delta z \approx dz$

	Calc I	Calc III
Differential	$dy = f'(x) dx$	$dz = f_x dx + f_y dy$
Meaning	Change in y along tangent line	Change in z along tangent plane
Approximation	$\Delta y \approx dy$	$\Delta z \approx dz$

Differentials give you the **change** in the output. Linear approximation gives you the **new value**. They're two sides of the same coin: $L = f + dz$.

Example 5: Differentials Approach

Example 5

Use differentials to approximate $f(2.05, 0.92)$ where $f(x, y) = \sqrt{25 - 4x^2 - y^2}$.

(Same problem as Example 4 — different method.)

Set $dx = 0.05$, $dy = -0.08$. Compute $dz = f_x dx + f_y dy$. Then $f(2.05, 0.92) \approx f(2, 1) + dz$.

Write the general differential:

$$dz = f_x(x, y) dx + f_y(x, y) dy$$

For $f(x, y) = \sqrt{25 - 4x^2 - y^2}$:

$$dz = \frac{-4x}{\sqrt{25 - 4x^2 - y^2}} dx + \frac{-y}{\sqrt{25 - 4x^2 - y^2}} dy$$

Identify the changes: $dx = 2.05 - 2 = 0.05$, $dy = 0.92 - 1 = -0.08$

Substitute at $(2, 1)$:

$$dz = (-2\sqrt{2})(0.05) + \left(-\frac{\sqrt{2}}{4}\right)(-0.08) = -0.1\sqrt{2} + 0.02\sqrt{2} = -0.08\sqrt{2}$$

Approximation: $f(2.05, 0.92) \approx f(2, 1) + dz$

$$= 2\sqrt{2} + (-0.08\sqrt{2}) = 1.92\sqrt{2} \approx 2.7153$$

Same answer as Example 4 — because it's the same idea!

Linear approximation says "plug in." Differentials say "find the change." Both give the same result.

Your Turn: Differentials Practice

Example 6

The volume of a right circular cone is $V = \frac{1}{3}\pi r^2 h$.

If the radius is measured as $r = 5$ cm and the height as $h = 12$ cm, each with a possible error of ± 0.1 cm, use differentials to estimate the maximum error in the calculated volume.

Find $dV = V_r dr + V_h dh$. Use $|dr| = |dh| = 0.1$ to get the maximum $|dV|$.

Write the general differential:

$$dV = V_r dr + V_h dh = \frac{2}{3}\pi r h dr + \frac{1}{3}\pi r^2 dh$$

Substitute $r = 5$, $h = 12$, $|dr| = |dh| = 0.1$:

$$|dV| \leq \frac{2}{3}\pi(5)(12)(0.1) + \frac{1}{3}\pi(25)(0.1) = 4\pi + \frac{2.5\pi}{3} = \frac{14.5\pi}{3} \approx 15.18 \text{ cm}^3$$

Homework

Section 14.4: 3, 9, 11, 21, 25, 27, 29, 33, 39, 43, 52